Projected Health Impact of WHO Cardiovascular Risk-Factor Targets, 2026–2050

WHO · University of Washington

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Study Overview

Aim 1: Research Questions

Central Question

What is the projected **burden-of-disease impact** of achieving the WHO cardiovascular risk-factor control targets by **2030**?

Sub-questions driving the analysis:

1. How many **cardiovascular events** (myocardial infarctions and strokes) could be averted?
2. How many **CVD deaths** could be prevented by 2030 and beyond?
3. How many **life-years** could be gained under full target attainment?
4. How would **YLLs, YLDs, and DALYs** change under joint achievement of:
   - 150M additional people with controlled hypertension
   - $\geq$50% lipid-lowering therapy coverage
   - 80% BP control among people with diabetes
   - 80% validated BP device availability

Scope

190 countries · Ages 20–95 · 2025–2050 ·

Intervention start: 2026

Model calibrated to 2019 baseline. First year of

Study Design at a Glance

Component	Detail
Model type	Discrete-time multi-state model (Well $\rightarrow$ Sick $\rightarrow$ Dead)
Calibration data	GBD 2023: incidence, case fatality, prevalence; all-cause & CVD-specific
Population data	UNWPP 2024 single-year age/sex projections
Intervention start	**2026** — first year intervention effects enter the projection
Scale-up period	Linear scale-up 2026 $\rightarrow$ 2030; rates held constant 2030–2050
Comparison	Business-as-usual: 2024 control rates, no additional scale-up
Outcome	CVD deaths averted vs. BAU; age-standardised mortality rate (ASMR)
Time horizon	2026–2050 (25-year horizon); model calibrated from 2019

Study Context

Objective

- Quantify CVD deaths avertable by achieving WHO 2030 cardiovascular targets
- Coverage: **190 countries**, 2025–2050
- Four causes: IHD, ischemic stroke, hemorrhagic stroke, hypertensive HD

Model

- Discrete-time state transition (Well $\rightarrow$ Sick $\rightarrow$ Dead)
- Calibrated to GBD 2023; projected with UNWPP 2024 populations
- 8 blood-pressure bins $\times$ GBD relative risks
- Baseline treatment effects explicitly de-duplicated

Research Question

What are the projected deaths averted from achieving WHO hypertension control and statin therapy targets across **190 countries**, 2025–2050?

Pending △

Confirm primary metric: deaths averted vs. DALYs vs. LYG

Data Sources

Data Sources and Key Variables

Source	Variables extracted	Role in model
GBD 2023 (IHME)	IR, CF, prevalence, background mortality (age/sex/country)	Baseline transition rates; initial states
GBD 2019 (IHME)	Relative risks per 10 mmHg SBP increase (IHD, stroke, HHD, by age)	Anchor BP-bin incidence; normalisation factor α
Ettehad et al. 2016	Trial effect sizes per BP category; diabetes-stratified	Treatment effect sizes by BP bin $\times$ cause
UNWPP 2024 (UN)	Population projections 2019–2050 (single-year age, sex, country)	Cohort sizes; rate denominators
WHO / NCD-RisC	Baseline HTN control rates (C_0); BP distribution by country	C_0; $P(\text{BP}_{\text{cat}})$
Country-specific	HTN control targets 2030 (Progress / Ambitious / Aspirational)	C_{target} per scenario

Scenarios

Four WHO Priority Targets

#	WHO Target	Modelled Scenario
1	150M more with HTN controlled by 2030	**HTN Control (General)**: 50% BP control by 2030
2	$\geq$50% eligible adults on lipid-lowering therapy by 2030	**Improved Statin Uptake**: 50% coverage by 2030
3	80% BP control among diagnosed diabetics by 2030	**HTN Control (Diabetes)**: 80% BP control in diabetics by 2030
4	80% availability of validated BP devices	Operationally embedded in HTN scenarios

Combination rule: $\varepsilon_{IR,\text{total}} = \varepsilon_{IR,\text{BP}} \times \varepsilon_{IR,\text{statin}}$

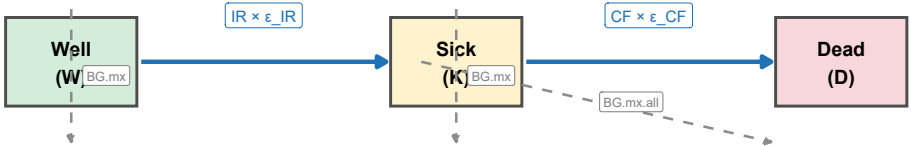
HTN Control Targets by 2030 — Country Distribution

[Figure: HTN target distribution -- data pending]

Mathematical Model

Model Architecture: Three Health States

Multi-State Transition Model | Annual time step | Ages 20-95 | 4 CVD causes



Key: $\varepsilon_{IR}, \varepsilon_{CF}$ = multiplicative effect ratios from the intervention module · BG.mx = background mortality · COVID-19 excess mortality applied 2020–2021 only

Intervention Module: 9-Step Pipeline

	Action	Output
1	Baseline BP distribution $P(\text{BP}_{\text{cat}})$ without treatment (rx $= 0$)	$P(\textit{BP}_{\textit{cat}} \mid a, s, t)$
2	GBD relative risks per 10 mmHg SBP $\rightarrow$ bin-specific incidence IR_{bin}	IR_{bin}
3	Diabetes-weighted Ettehad trial effect sizes E_{trial} per BP bin $\times$ cause	E_{trial}
4	Linear coverage scale-up from C_0 to C_{target}, **starting 2026**	$C_t(t,\ \textit{BP}_{\textit{cat}})$
5	Coverage adjustment: $E_{\text{adj}} = E_{\text{trial}}(C_t - C_0) \,/\, (1 - E_{\text{trial}}\, C_0)$	$E_{adj}(t)$
6	New bin incidence: $IR_{\text{bin,new}} = IR_{\text{bin}} \times (1 - E_{\text{adj}})$	$IR_{bin,new}$
7	Aggregate: $IR_{\text{new}} = \sum IR_{\text{bin,new}} \times P(\text{BP}_{\text{cat}})$	IR_{new}
8	Incidence effect ratio: $\varepsilon_{IR} = IR_{\text{new}} \,/\, IR_{\text{original}}$	$\varepsilon_{IR} \rightarrow$ *model*
9	Case-fatality effect ratio: $\varepsilon_{CF} = 1 - \rho_{CF} \times C^{\Delta}_{\text{agg}}$	$\varepsilon_{CF} \rightarrow$ *model*

Blue row = intervention start (2026). Green rows = outputs passed to state-transition model.

Key Equations

Incidence pathway

Step 2 — GBD RR anchors bin incidence:

$$IR_{\text{bin}} = \frac{RR_{\text{GBD}} \times IR}{\alpha}, \quad \alpha = \sum_{\text{cat}} P_{\text{cat}} \cdot RR_{\text{GBD}}$$

Step 5 — baseline-adjusted effect (no double-counting):

$$E_{\text{adj}}(t) = \frac{E_{\text{trial}}\,(C_t - C_0)}{1 - E_{\text{trial}}\,C_0}$$

Steps 7–8 — population-weighted effect ratio:

$$\varepsilon_{IR} = \frac{\sum IR_{\text{bin,new}} \cdot P_{\text{cat}}}{IR_{\text{original}}}$$

Case fatality pathway

Incremental coverage above baseline:

$$\Delta C_\delta(t) = \text{max}(C_t - C_0,\ 0)$$

Aggregate across hypertensive bins (≥140 mmHg):

$$C^{\Delta}_{\text{agg}} = \frac{\sum_{\geq 140} \Delta C_\delta \cdot P_{\text{cat}}}{\sum_{\geq 140} P_{\text{cat}}}$$

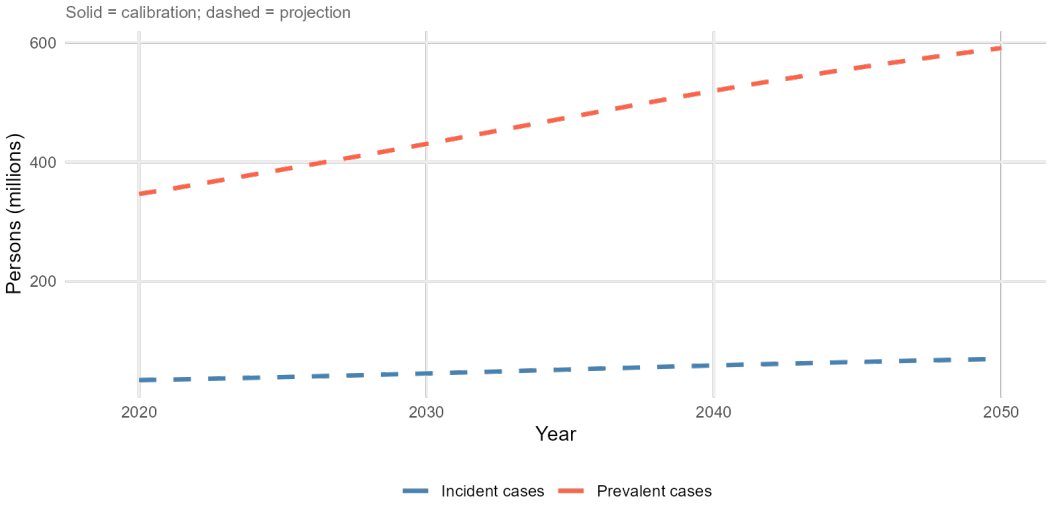
CF effect ratio:

$$\varepsilon_{CF} = 1 - \rho_{CF} \times C^{\Delta}_{\text{agg}}$$

Cause	ρ_{CF}
IHD	0.24

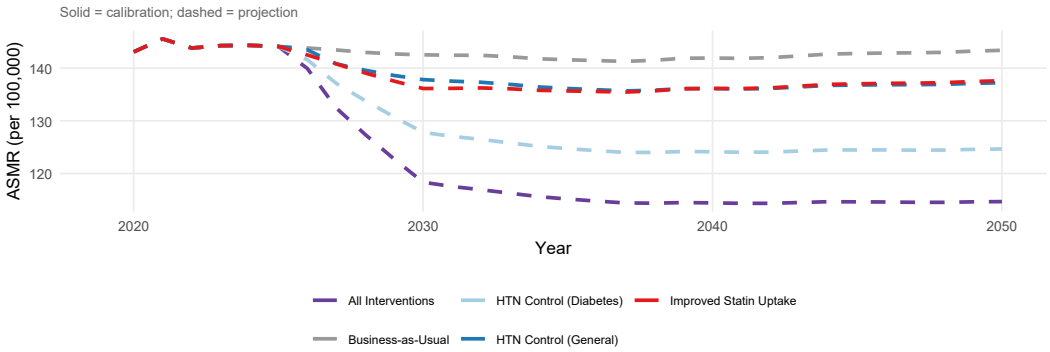
Results

Baseline Disease Trends (Business-as-Usual)



Incident and prevalent CVD cases 2019–2050; solid = calibration, dashed = projection.

Age-Standardised CVD Mortality Rate



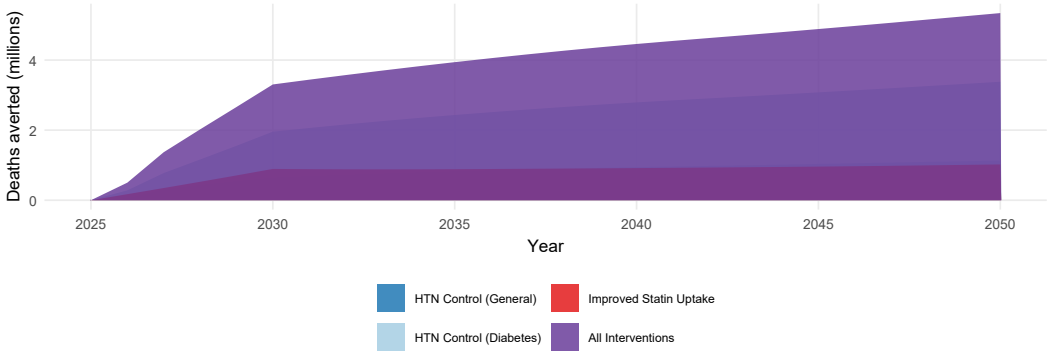
Summary: Cumulative Deaths Averted, 2025–2050

Scenario	Cumulative (M)	Avg./year (M)	% of BAU deaths
HTN Control (General)	20.7 M	0.8 M	3%
HTN Control (Diabetes)	61.4 M	2.5 M	8.8%
Improved Statin Uptake	21.3 M	0.9 M	3.1%
All Interventions	**99.0 M**	**4.0 M**	**14.2%**

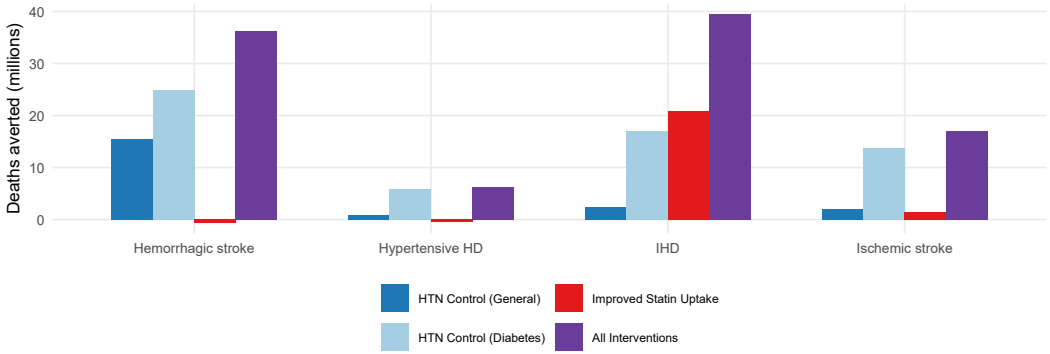
Headline

All Interventions: **99.0 million deaths averted** globally, 2025–2050 ($\approx$ 4.0 M/year).

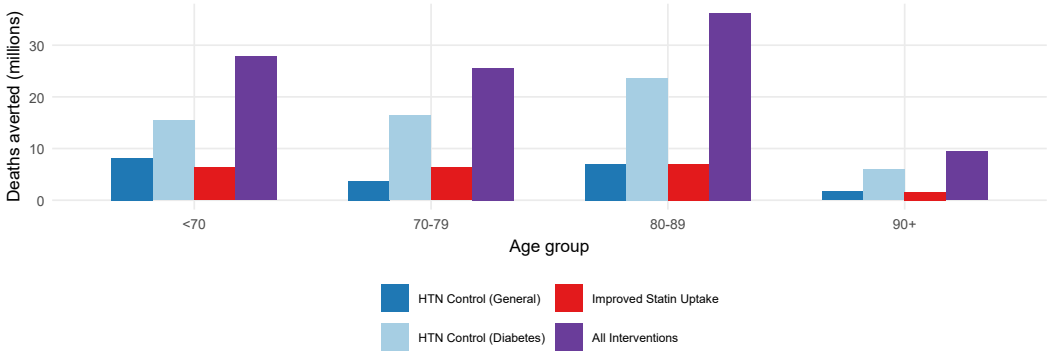
Annual Deaths Averted Over Time



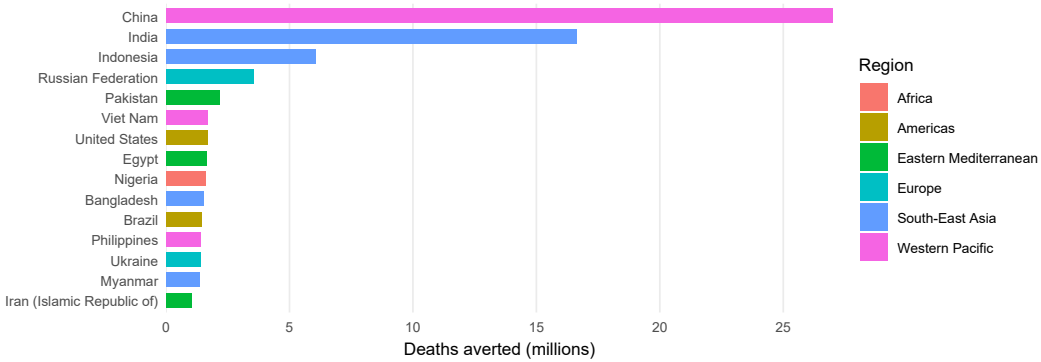
Deaths Averted by Cause of Death



Deaths Averted by Age Group



Top 15 Countries by Projected Impact



Regional Summary

Region	BAU deaths (M)	Averted (M)	% averted
Western Pacific	227.3	32.6	14.3%
South-East Asia	175.8	27.4	15.6%
Europe	117.2	13.6	11.6%
Africa	48.7	9.6	19.8%
Eastern Mediterranean	64.3	9.1	14.2%
Americas	62.8	6.7	10.7%

Observation

Absolute gains largest in high-burden regions; proportional gains reflect current coverage gaps.

Discussion

Interpretation

Key findings

- **All Interventions** yields largest gains: HTN control + statins are complementary
- **IHD** and **ischemic stroke** drive the majority of avertable deaths
- Gains **accelerate after 2030** as coverage compounds
- **Older adults (70+)** benefit most in absolute terms
- **Top 15 countries** account for >75% of avertable burden

Policy Implication

Combined WHO Targets 1 + 2 are synergistic — neither alone captures the full benefit.

Uncertainty △

Effect sizes from Ettehad (2016). Updated NMA estimates pending.

Limitations

Model structure & assumptions

- Stable transition rates assumed; no dynamic feedback between states
- Static BP distribution (log-normal): no secular trend and no endogenous updating as coverage increases
- No heterogeneity in adherence: newly covered individuals assumed fully adherent
- COVID-19 excess mortality only modelled for 2020–2021; long-term effects excluded

Data & parameters

- RCT-based effect sizes (e.g., Ettehad ρ_{CF}): real-world effectiveness at scale may be lower
- GBD 2023 uncertainty intervals not propagated
- Baseline CVD from NCD-RisC: measurement

Scenarios & implementation realism

- Linear scale-up assumed (2026–2030); real programmes may follow non-linear trajectories
- No explicit health-system capacity constraints modelled
- No equity dimension: impacts not stratified by socioeconomic position (e.g., income quintile)
- Interactions with statins, diabetes care, and BP device availability not included in Aim 2; incorporated in Aim 1 combined scenarios

Next Steps

Analysis priorities

1. Probabilistic sensitivity analysis on effect sizes and coverage
2. Add life-years gained (LYG) as co-primary outcome
3. Stratify by income group (World Bank)
4. Validate against observed trends (2019–2023)
5. Extend to Aim 2: cost-effectiveness and budget impact

Pending Decisions △

- Primary metric: deaths vs. DALYs vs. LYG?
- PSA parameter scope?
- Country inclusion threshold?
- Headline period: 2025–2030 or 2025–2050?
- Journal target?

Summary

1. The gains are large and achievable

All-interventions: **99.0 million deaths averted** globally, 2025–2050.

2. HTN control + statins are synergistic

Combined deployment outperforms the sum of either alone.

3. A small number of countries drive most of the gain

Targeting the top 15 countries could unlock >75% of the global avertable burden.

Meeting WHO 2030 CVD targets is not just aspirational — it is quantifiably lifesaving.

Appendix

Mathematical Notation Reference

Symbol	Description
$IR_{a,s,c}, CF_{a,s,c}$	Baseline incidence & case-fatality rates
$\varepsilon_{IR}, \varepsilon_{CF}$	Effect ratios applied to baseline rates
$C_{\text{baseline}}, C_{\text{target}}$	Baseline & target control rates
ΔC_{target}	Incremental coverage to achieve
RR_{GBD}	GBD relative risk per 10 mmHg SBP
E_{trial}	Ettehad trial effect size per 10 mmHg
w_{DM}	Diabetes prevalence weight (default 0.10)
W, K, D	Well, Sick, Dead population states
$T_{\text{start}}, T_{\text{end}}$	Scale-up start (2025) & end (2030)

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- **UNWPP** (2024). World Population Prospects 2024. United Nations.
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Thank You

Questions & Discussion

Contact: David Watkins | davidaw@uw.edu | University of Washington